

Antisense sequence choices for extraskeletal myxoid chondrosarcoma

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Abstract

We present a source-qualified catalogue of 215 antisense oligonucleotide designs for 43 extraskeletal myxoid chondrosarcoma fusion junctions. We compare five binding positions per junction against mature RNA sequences from seven fusion-parent genes. Including additional normal RNA variants changes the preferred positions under the primary matching rule at nine junctions. Only five junctions match deposited fusion sequences; two have reference reconstructions and 36 remain hypothetical. The catalogue supports sequence-based experimental planning, but does not identify a biologically selective lead.

Keywords: extraskeletal myxoid chondrosarcoma; antisense oligonucleotide; fusion junction; RNA isoform

Main text

We developed a catalogue of antisense oligonucleotide (ASO) sequences for extraskeletal myxoid chondrosarcoma (EMC), with the evidence behind each fusion junction made explicit. Including additional normal RNA variants changes the positions favored by our primary matching rule at nine of 43 junctions. Fusion-junction ASOs have been tested experimentally in other cancers,[1] and sequence enumeration and off-target screening are established methods.[2] Our contribution is an EMC-specific, traceable comparison that shows how the underlying fusion and normal RNA sequences affect design choices.

We generated five 16-base ASOs per junction by moving the binding site one base at a time. Each site includes six to ten bases from the upstream fusion partner. The proposed chemistry places a six-base DNA region between two five-base locked-nucleic-acid regions. We compared all 215 designs with 78 distinct mature RNA sequences from the seven parent genes. Each design was scored by its longest uninterrupted normal-RNA match spanning the entire DNA region. We first favored the lowest score. Among ties, we favored the design whose closest normal 16-base sequence had the most differences. All remaining ties were retained.

Including additional RNA variants, while keeping the seven genes fixed, lengthened these uninterrupted matches for 23 designs. At the deposited TCF12-NR4A3 junction, an evenly split binding site initially appeared preferable. An alternative normal NR4A3 transcript increased its match from seven to eleven bases. Moving the site by one base then became preferable under our rule (Table 1). The complete catalogue contains one final choice at 27 junctions and tied choices at 16; it includes every tested sequence and comparison, not just the selected designs.

The fusion evidence is equally consequential. Seven deposited sequences support five distinct junctions. Two further junctions can be reconstructed from reported coordinates or an engineered-construct description; the other 36 are hypothetical reference joins. For the published USZ20-EMC1 model,[3] reconciling transcript identifiers and coordinates supports a different reference junction from a literal reading of its exon labels. This remains a reference reconstruction, not a recovered patient consensus. The second model, USZ22-EMC2, remains unresolved.

These are choices under a declared sequence rule, not experimentally established ASO rankings. The comparison covers seven parent genes, not the whole transcriptome or unspliced RNA. It also omits accessibility, delivery and cleavage measurements. Sequence differences alone do not establish biological selectivity,[4] and the proposed chemistry remains untested. The practical output is therefore a source-qualified catalogue for planning experiments: it identifies plausible binding sequences, exposes sensitivity to normal RNA variants, and distinguishes documented junctions from assumptions. Supplementary data provide the complete designs, evidence, normal sequences and reproducible analysis.

Table 1. Including normal RNA variants changes the preferred position at the deposited TCF12-NR4A3 junction.

Upstream bases	One RNA per gene: match (bases)	With RNA variants: match (bases)	Fewest differences (expanded comparison)
6	11	13	3
7	8	12	4
8	7	11	4
9	8	8	5
10	8	8	4

Match length is the longest exact normal-RNA match spanning the central six DNA bases; shorter is favored. The final column is the fewest differences from any complete normal 16-base window; more is favored among ties. The last column uses the expanded comparison. Upstream bases come from TCF12. All comparisons use the same seven genes. Bold marks the final choice: ASO 5'-GGCATATCCATCAGAT-3'.

Statements and Declarations

Ethics approval

Not applicable; this computational study used published sequence and reference records, with no new recruitment or specimen collection.

Consent to participate

Not applicable; no participants were recruited.

Consent for publication

Not applicable; no identifying patient information is presented.

Declaration of conflicting interests

The author declares no competing interests.

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Data availability

Complete tables and frozen inputs accompany the supplement. Reproducible analysis is available at github.com/trimcrae/Rare-cancers, commit a916dab2979e27f930b417243f021b6c2b2ca371, under research/release-candidates/PUB-ASO/2026-09-30-full-catalogue.

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